

A Jai-inspired systems language in Rust. Incremental salsa front end, lossless CST, typed SSA mid-end, two execution engines held byte-identical by a differential harness.

6/6 gates green 924 tests ADR-0066 latest W3 complete · W4 comptime next

TESTS	CORPUS	ADRS	DIAGNOSTICS	EDITOR CHECKS
924	149	66	88	156
workspace, all passing	jr files, both engines	0001 to 0066, immutable	codes, E0258 next free	Neovim, verified not gated

The vertical slice, and what it costs to be honest about it

EXIT CRITERIA DONE hello.jr runs in the VM DONE compiles to a native binary DONE rustc-grade diagnostics DONE formatter round-trips the corpus DONE editor integration (Neovim) OPEN verified Linux x86-64 CI run The last one needs a push. Configured, never run, so it is a decision rather than a technical gap.	HOW CORRECTNESS IS ESTABLISHED Both engines run every corpus program and must agree, on output and on trap wording, from one shared formatter. The differential harness compares a failing program too. A trap's whole stderr — reason, location and now the call chain — must match byte for byte, which is the only thing holding two very different mechanisms together: the VM pushes a <code>ProcRef</code> and resolves names from the HIR, while native writes name pointers into a mutable global its generated helper walks at trap time. It found a real asymmetry. <code>main</code>'s frame is pushed by its caller, and native's caller is the C entry shim while the VM's is <code>run_main</code> — so native printed one frame fewer until the shim learned to push it. Nothing but the byte comparison would have noticed. An ADR can be wrong about its own example. This wave's draft predicted two callees would both be inlined away; running it showed only leaves inline, so one survived with a real frame. The forks were sound, the illustration was not — and running the example while writing the ADR, not after, is what caught it.
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The language today

WORKS, END TO END Full integer tower, bool, string, pointers float32 and float64, plain IEEE-754, no traps struct, nominal, one level union, nominal, untagged, fields at offset 0 enum and enum_flags, namespaced, bare dot-member Fixed arrays and views, bounds-checked, returnable cast and xx from context; operator overloading Trapping arithmetic, wrapping variants, bitwise if, else, while, for, break, continue, defer, using Multiple returns, named args, literal defaults import, foreign, system_library, #scope_module One trivial compile-time run, folded Traps name their source line and the live call chain Bounds checks, strippable by build setting or #no_abc context, a hidden parameter passed by pointer Procedures as values: call, pass, return, struct field null, a context-typed pointer literal; malloc and free An allocator in the context, installed and called through Pointer offset <code>p + n</code>, <code>n + p</code>, <code>p - n</code> — element-scaled, unchecked push_context, a block with its own copy of the context talloc / reset_temporary_storage — a per-context bump arena	ABSENT (WAVE) pointer difference <code>p - q</code>, deferred percent on floats, <code>is_nan</code>, math (W7) a tagged variant type (W4.5) explicit backing type; a switch (W4.5) array literals, sub-slicing unary, index and call overloading transmute; float printing #must; a multi-result call in a return polymorphs, macros (W5) arbitrary run, RTTI, insert (W4) a per-frame line; inlined frames (none exist) a per-index #no_abc; any other build setting a cross-file or foreign proc value; comparing one cast to a pointer the difference <code>p - q</code>; <code>p[n]</code>; ordering hands out *u8 only; alignment; a growable region
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No error-handling model yet — ADR-0008 reserves the slot and the first half exists (several return values); #must is owed its own ADR. No GC and no RAIi, which is a design value rather than a gap. A union is untagged, so reading a field other than the one last written reinterprets bits: documented in three places because it cannot be diagnosed. No indirect calls, which is the single largest gap in the project and is what blocks the rest of W3.

Compiler internals, 17 crates

STAGE	STATE	NOTE
Lexer, parser, CST, typed AST	works	Hand-written, error-recovering, trivia-preserving
Formatter	works	Pure function over the CST; lost source in 7 of the last 8 waves
HIR, name resolution, modules	works	Flat import merge; cycles legal; export filtering
InternPool: types, values, layout	works	One layout computation and one integer evaluator, shared
Sema: signatures, checking	works	E0212 to E0257; no const-eval here, by design
MIR: typed SSA	works	Block parameters, notphis; explicit bounds check and zeroing
Mid-end	5 passes	Inline, forwarding, const-prop, DCE, plus the bounds-check strip
Const-eval	works	Runs MIR through the bytecode VM
VM: register bytecode, libffi	works	No JIT; indirect calls, and malloc from its own region
Cranelift back end	works	Aggregate returns via sret; indirect calls via func_addr
LLVM back end	not started	W8 owns it
Language server	12 caps	Diagnostics, hover, goto, completion, rename, actions, hints
Neovim integration	works	Runtimepath dir, no plugin manager; 156 checks
Driver	stub	Should consume the workspace notion that now exists

Roadmap: W1 and W2 closed, W3 started

W1 Data	done	Numeric tower, wrapping ops, enum, enum_flags, union, arrays, views, cast, xx, operator overloading. Closed by ADR-0048.
W2 Flow and scope	done	for with it and it_index, labelled break, defer, using, multiple returns, named and default arguments, #scope_module. Closed by ADR-0054; ADR-0055 then closed a gap six corpus files had carried for eleven waves.
W3 Runtime core	done	context (ADR-0057), the bounds-check build setting (ADR-0058, finishing ADR-0003), indirect calls (ADR-0059), null plus a memory source (ADR-0060/0061), the allocator protocol (ADR-0062), push_context (ADR-0063), pointer arithmetic (ADR-0064), temporary storage (ADR-0065), and traps with backtraces (ADR-0066) — a trap names the frames that were live, byte-identically in both engines. Closed by ADR-0066; a source-level backtrace with inlined frames is deferred, because inlined frames have no runtime existence.
W4 Comptime	not started	Arbitrary compile-time execution, RTTI and Type values, insert, code. PLAN section 5 names this the project's top risk: sema and comptime become mutually recursive.
W4.5 Pattern matching	not started	switch with exhaustiveness, a bare dot-member as a case, and a tagged variant type. Was missing from the wave table entirely — two accepted ADRs deferred decisions to it while no wave scheduled it.
W5 Polymorphism	not started	Polymorphic parameters, modify, bake_arguments, expand macros with hygiene, instantiation caching.
W6 Metaprogram	not started	Workspaces, the compiler message loop, build scripts replacing makefiles, note attributes.
W7 Stdlib	not started	Written in Jairs: String, dynamic array, hash table, Sort, Math, Random, File.
W8 Performance	not started	LLVM via inkwell for release builds, inliner maturity, struct-of-arrays, SIMD, parallel sema. Also owns the compile-throughput number, still unpublished.
W9 Tooling depth	not started	Semantic tokens are the one LSP capability still missing; the other twelve landed early.
W10 Graphics, in Jairs	not started	Window creation through foreign calls, Metal then Vulkan, immediate-mode 2D.

What this session did, and how each wave was scoped

EIGHTEEN WAVES SHIPPED	TWO CLAIMS THE CODE DISPROVED
ADR-0049 through 0066: for and defer, using, aggregate returns, multiple returns, named and default arguments, scope visibility, imported constants, float constants, context, the bounds-check build setting, indirect calls, null plus a memory source, the allocator protocol, push_context, pointer arithmetic, temporary storage, and trap backtraces.	An ADR corrected mid-wave. ADR-0060 §4 asserted the VM dereferences a malloc'd pointer via libffi. Running the corpus file the same ADR promised would pass disproved it: the VM's memory is a linear region and a host address is not an offset into it. ADR-0061 corrects it — the VM allocates from its own region, native calls libc, the bits differ and nothing observes them.
Test count 900 to 924. Corpus 116 to 149 files. Neovim checks 103 to 156.	A diagnostic that could not be acted on. Installing an imported #foreign procedure into an allocator field reported “expected (s64) -> *u8, found (s64) -> *u8” — the same text twice, because the types differ only in an invisible calling convention. It is E0256 now, the code that says to wrap it.
W2 and W3 both closed: an allocator in the context, push_context to scope it, raw pointer arithmetic, a temporary-storage arena built from all three, and a trap that names the frames that were live. W4 — comptime, the project's top risk — is next.	

Each wave was scoped by writing the feature and seeing what the compiler refused, not by reading the handoff. That found gaps the handoff had missed in both directions: a proc-pointer struct field it called absent (it worked), and a void-returning proc-pointer type it never mentioned (unspellable, and blocking).

THE THING WORTH YOUR DECISION

Fourteen waves sit uncommitted across five branches. Two waves ago a careless git checkout reverted the tree-sitter grammar nine waves and cost an hour's reconstruction — the concrete price of not committing. Committing each wave as it goes green would bound the damage from any future slip to a single wave.

Recorded as a recommendation, not a change: the authorisation is yours to give.

Sources: PLAN.md §1.5 and §7, the ADR directory, docs/decisions/DECISIONS.md, and the six gates run today. Every number was measured rather than carried forward — the test count from a full workspace run, the corpus count from a file walk, the diagnostic codes from the const E0nnn definitions across the crates, and the editor-check count from a real verify.lua run. Rebuild with typst compile jairs-dashboard.typ.